Biology by the numbers - Exercise 1

Submit your answers by uploading a PDF to the HW 1 Folder. You can submit scanned hand-written answers if your handwriting is legible. Be honest with yourself about your handwriting. Here and in the future, any added insights or suggestions might earn you a bonus.

1. Read *Why should we care about the numbers?* (pg xxiii) through Order-of-magnitude biology toolkit (ending pg xxx).

2. Compose a back-of-the-envelope Fermi style estimate on any question you wish (doesn't have to be biological). Write out the question and your calculation. Do not look up numbers from the web, rely on your intuition. You don't have to get the "correct" number!

3. *You are not alone!*

Part a)

(i) Justify that assuming a surface area of 6 μm^2 and a volume of 1 μm^3 for a characteristic bacterial cell (e.g. *E. coli*) is reasonable (use the accompanying figures).

(ii) Express this volume in femtoliters.

(iii) Make an estimate of the mass of such a bacterium.

Part b)

Try to make an estimate of the total number of bacteria inhabiting your intestine (where most of the bacteria in the body are thought to reside).

(Part c)

Estimate the total number of human cells in your body.

(Part d)

Using the above, calculate the ratio of bacteria to human cells in the body.

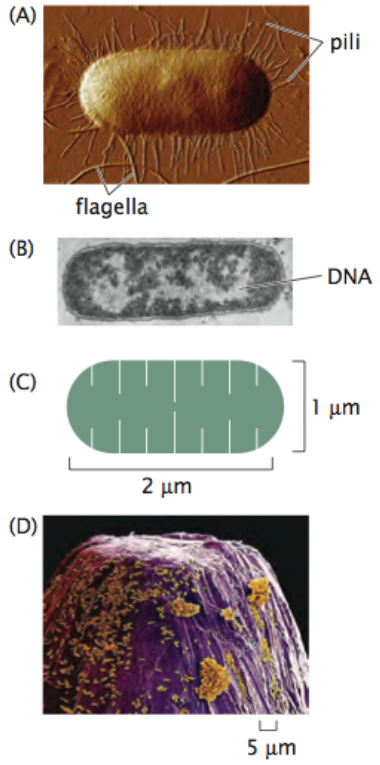

Figure 2.1: *E. coli* as a standard ruler for characterizing spatial scales. (A) Atomic-force microscopy (AFM) image of an *E. coli* cell, (B) electron micrograph of a sectioned *E. coli* bacterium, and (C) the *E. coli* ruler. (D) Bacteria on the head of a pin giving an impression of how our standard ruler compares with the dimensions of everyday objects. (A, courtesy of Ang Li; D, courtesy of Tony Brain.)

4. As the name of our course suggests, we care a lot about numbers. One part of this passion involves the correct way for reporting significant digits when writing numbers. Please read the following chapter in the book "RIGOROUS RULES FOR SLOPPY CALCULATIONS" (pg xxxii) and revisit your answers to questions 2 and 3 based on these guidelines.